

\$123.2M

Total Revenue Sa...

\$31.5K

Total Orders

\$19.8K

Total Customers

\$3.9K

Avg Order Value

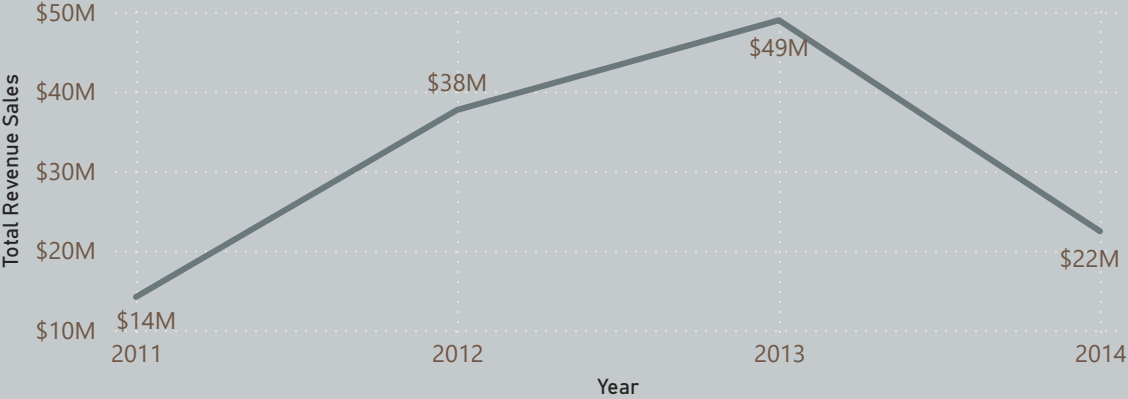
\$12.8M

Profit

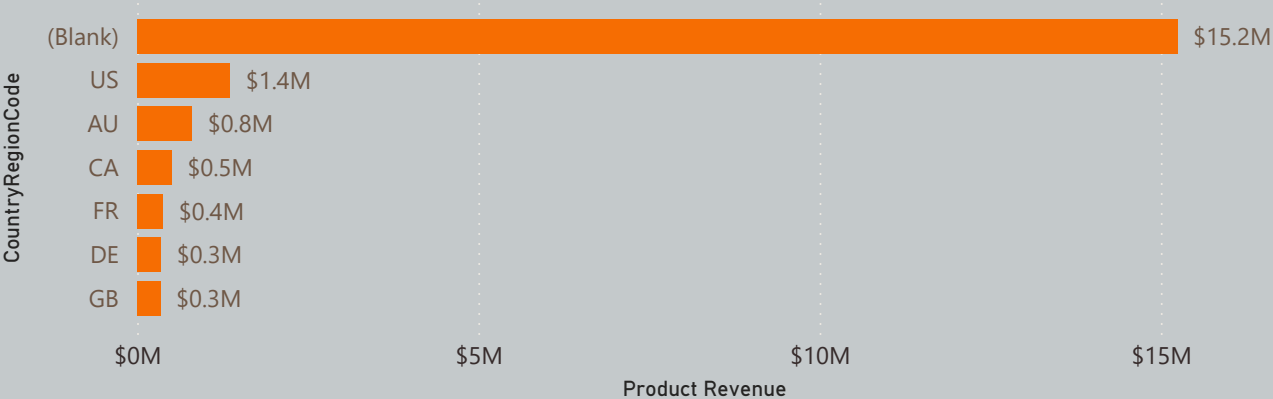
11.7%

Profit Margi...

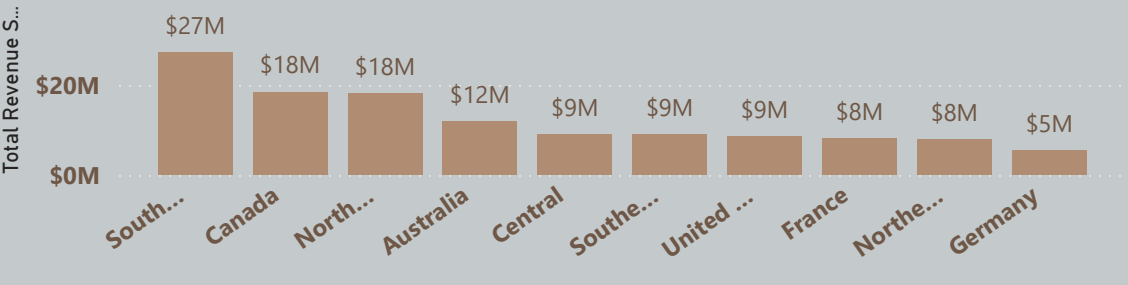
Total Revenue Sales by Year



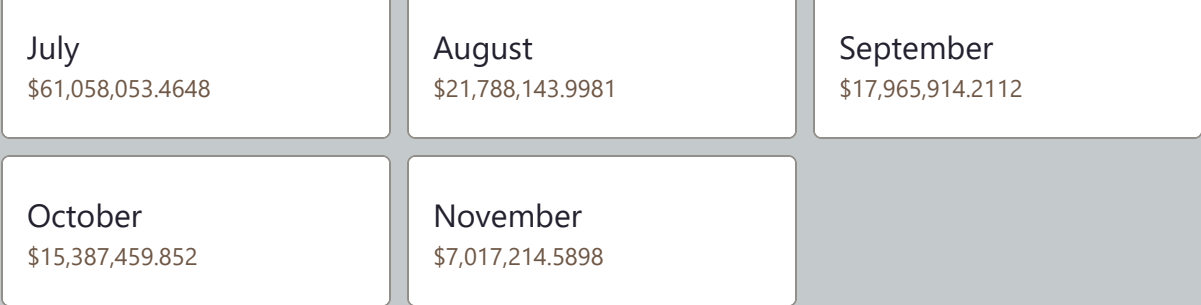
Product Revenue by Territory



Total Revenue Sales by Name



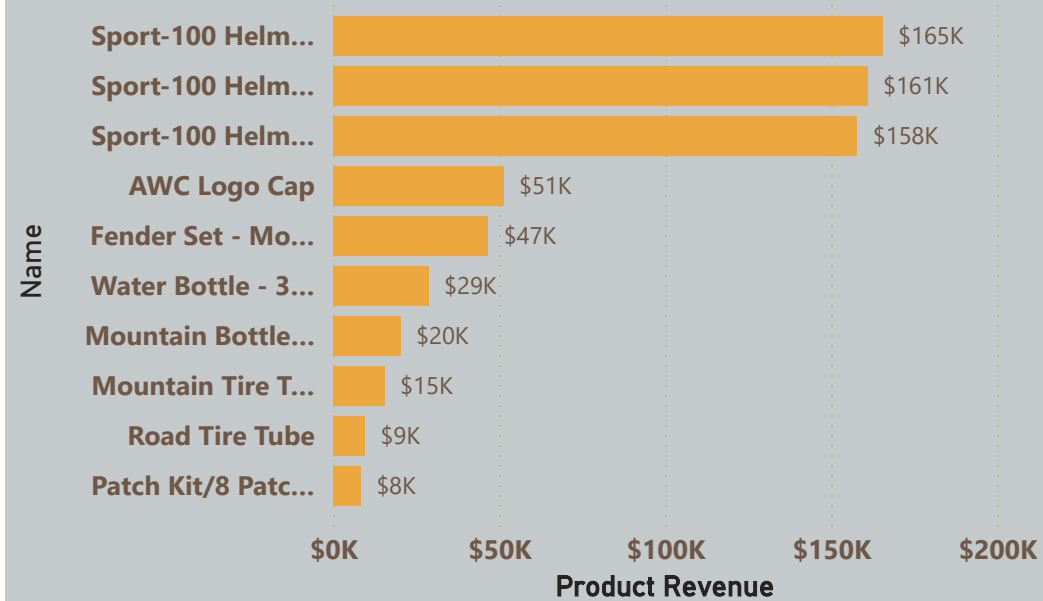
Total Revenue Sales by Month



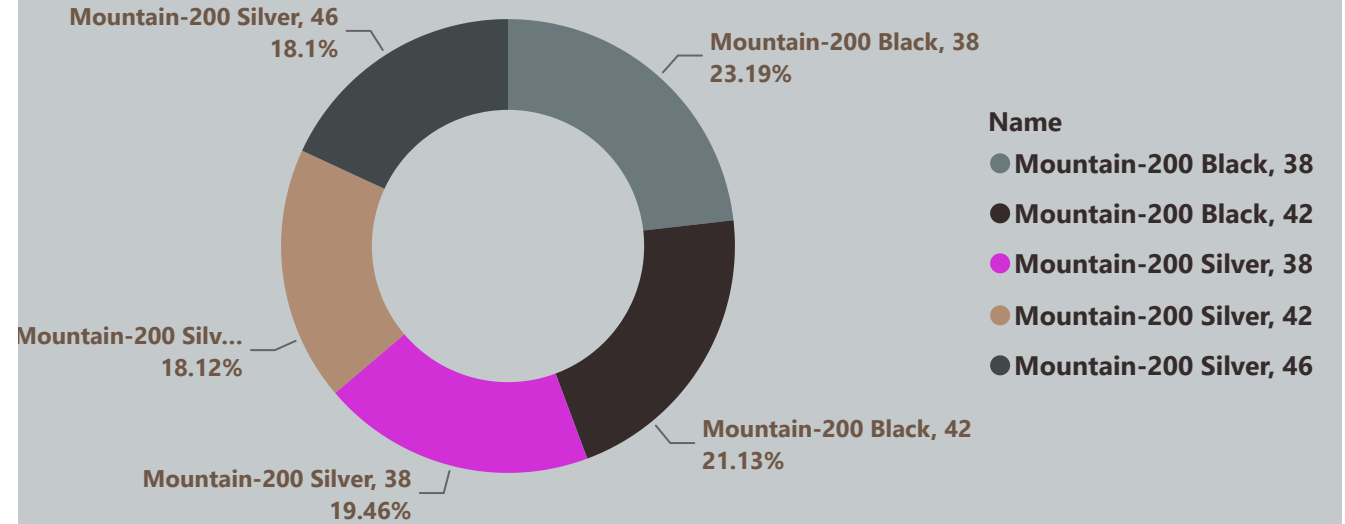


product performance

Top product chart



Product Revenue by Name



Product Revenue and Total Quantity by Name



CUSTOMER ANALYSIS DASHBOARD

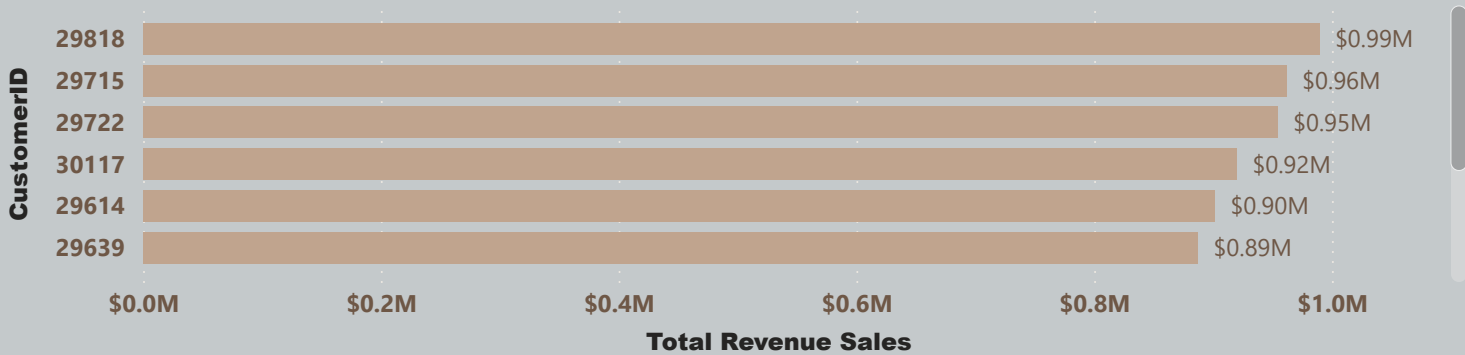
\$19.8K

Total Customers

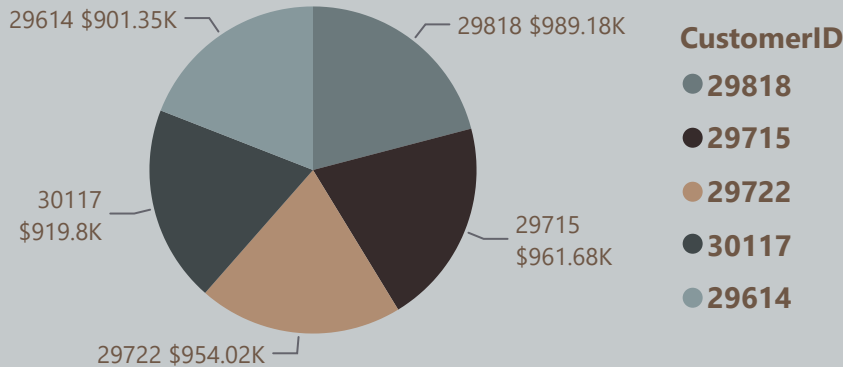
\$6.2K

Avg Revenue per Custom...

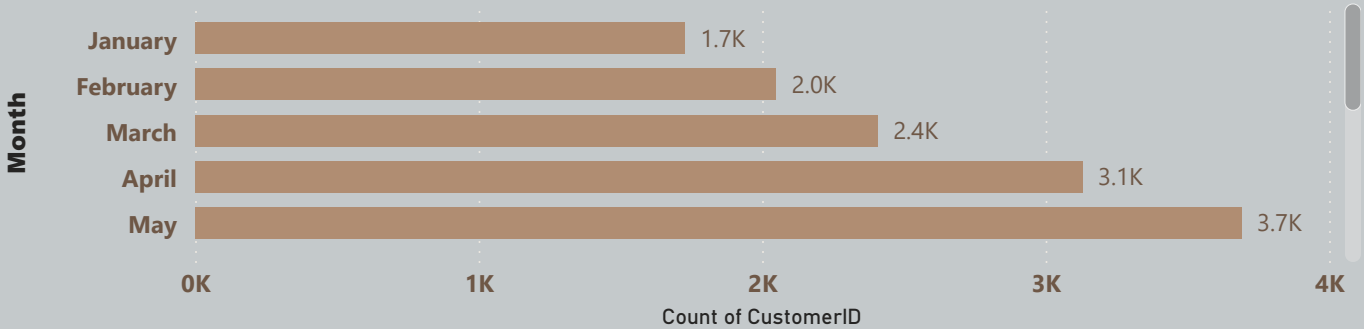
Total Revenue Sales by CustomerID



Total Revenue Sales by CustomerID



Active Customers by Month

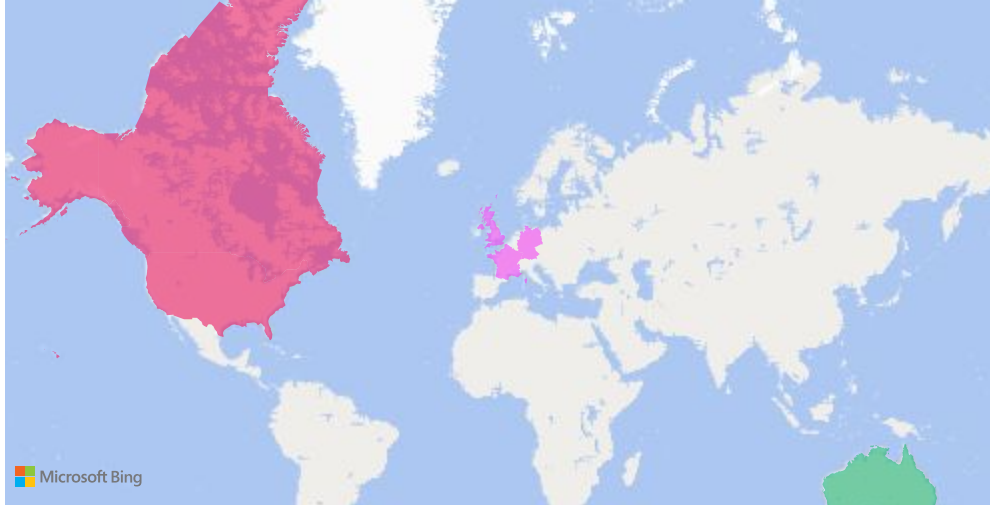




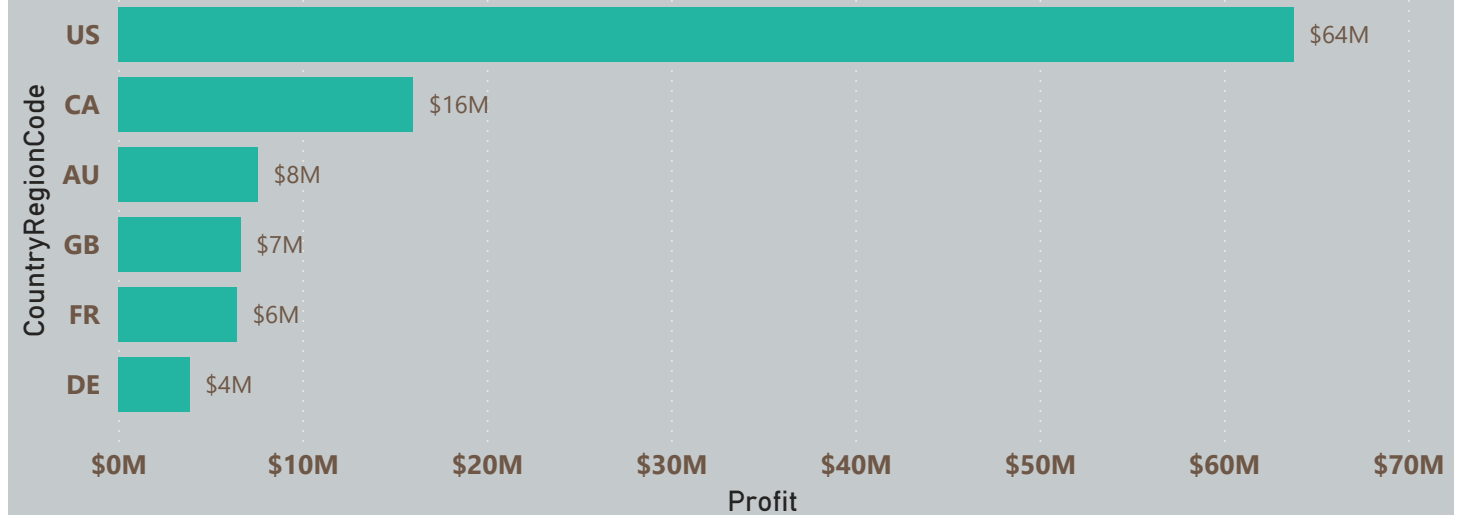
Territory Sales Insights

Total Revenue Sales by CountryRegionCode and Group

Group ● Europe ● North America ● Pacific

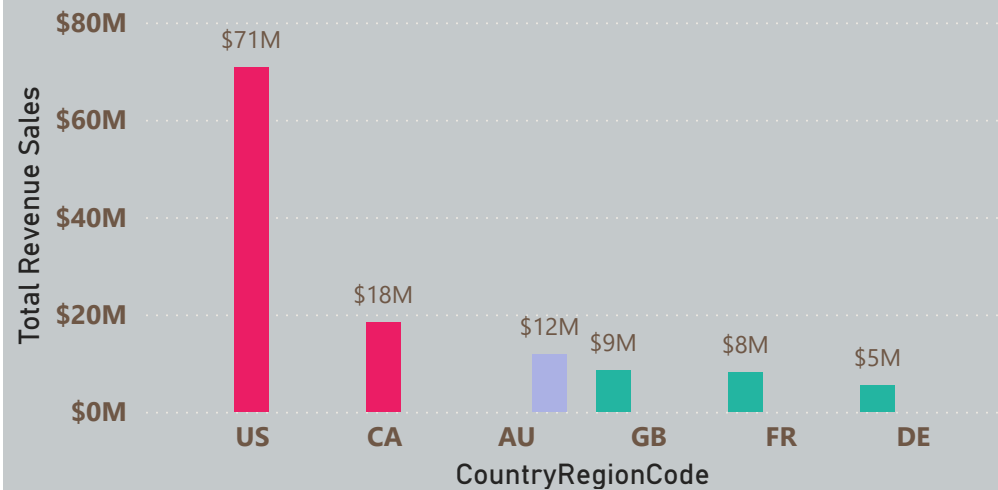


Profit by CountryRegionCode



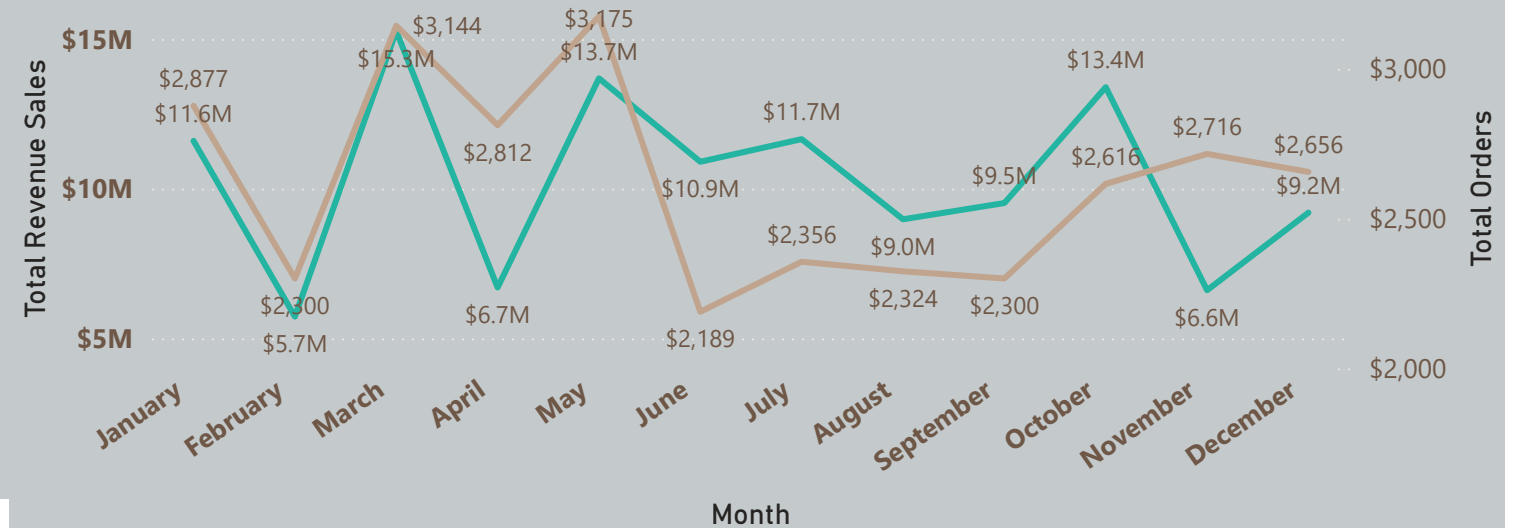
Total Revenue Sales by CountryRegionCode and Group

Group ● Europe ● North America ● Pacific



Total Revenue Sales and Total Orders by Month

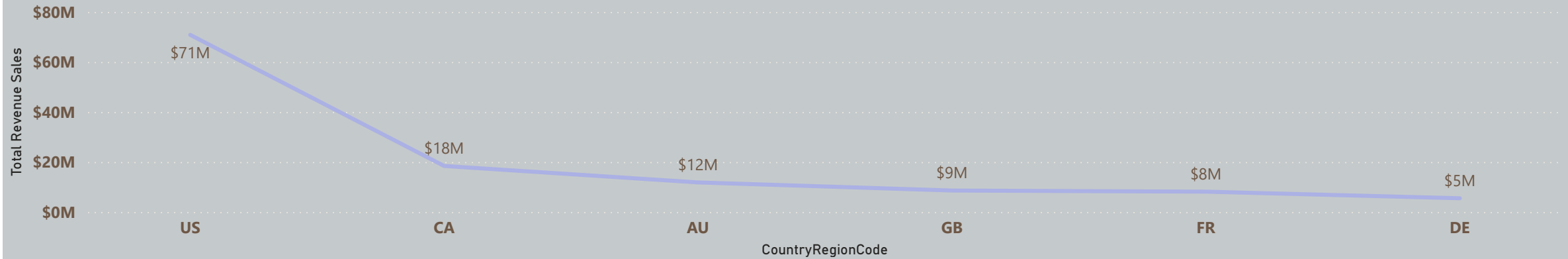
● Total Revenue Sales ● Total Orders





INSIGHTS PAGE

Sales Trend



✦ REVENUE & PRODUCTS

- Revenue is heavily concentrated in a small group of **top-performing products**, indicating a strong dependency on high-demand items
- Lower-performing products contribute minimally, suggesting potential for product pruning or marketing improvement
- Product mix is uneven, with a few categories driving most of the overall sales

✦ REGIONAL PERFORMANCE (TERRITORY INSIGHTS)

- Certain territories consistently outperform others in total sales contribution
- High-performing regions show stable and growing demand patterns
- Underperforming regions may require improved marketing, distribution, or customer engagement strategies

✦ CUSTOMER BEHAVIOR

- A small group of customers contributes a significant portion of total revenue (high-value customers)
- Customer purchase patterns suggest repeat buying behavior in key segments
- Customer retention appears stable, indicating strong loyalty within the customer base

📈 ✦ SALES TRENDS (TIME INSIGHTS)

- Sales trends show clear **seasonal fluctuations**, with predictable peaks and dips across the year
- Overall sales trend indicates either steady growth or cyclical movement depending on the period analyzed
- Periodic spikes suggest influence of promotions, demand cycles, or external market factors

🧠 ✦ BUSINESS OVERALL INSIGHT (IMPORTANT FOR MARKS)

- Revenue performance is driven by a combination of **product concentration + regional strength**
- Business growth depends heavily on optimizing top products and expanding weaker regions
- Data suggests opportunities for better diversification across products and territories

Nkwonta Chimamanda

ALT/SOD/025/5006

JUNE 21 2026

ADVENTUREWORKS CYCLES – SALES ANALYSIS REPORT

1. Introduction

This report analyzes AdventureWorks Cycles using the 2022 dataset, focusing on sales performance, product performance, customer behavior, and territory analysis. The objective is to generate actionable insights to support data-driven decision-making and improve overall business performance.

2. Data Preparation and Exploration

The dataset was provided as multiple Excel files, which were extracted and imported into Power BI. Key tables used include Sales SalesOrderHeader, Sales SalesOrderDetail, Sales Customer, Production Product, Production ProductSubcategory, Production ProductCategory, and Sales SalesTerritory.

Data cleaning was performed by removing duplicates and excluding missing values to improve accuracy. Each table was explored to understand its structure and relevant fields.

3. Data Modeling

Relationships were created to connect tables using primary and foreign keys:

- SalesOrderHeader ↔ SalesOrderDetail (SalesOrderID)
- SalesOrderHeader ↔ Customer (CustomerID)
- SalesOrderDetail ↔ Product (ProductID)
- Product ↔ ProductSubcategory ↔ ProductCategory
- SalesOrderHeader ↔ SalesTerritory (TerritoryID)

This enabled accurate aggregation and cross-table analysis.

4. Measures and Calculations

Key performance indicators (KPIs) were created using DAX:

- Total Revenue (SUM of TotalDue)
- Product Revenue (SUM of LineTotal)
- Total Orders (Distinct count of SalesOrderID)
- Total Customers (Distinct count of CustomerID)
- Average Order Value (Revenue ÷ Orders)

To improve analysis, additional metrics such as Profit and Profit Margin were derived. Time-based analysis was performed using OrderDate to identify trends.

5. Dashboard Development

A five-page interactive dashboard was developed:

- **Executive Dashboard:** Displays KPIs, revenue trends, and sales by territory
- **Product Performance:** Shows top products, category contribution (donut chart), and revenue vs quantity
- **Customer Analysis:** Highlights top customers and revenue contribution
- **Territory Performance:** Uses maps and charts to compare regional performance globally
- **Insights Page:** Combines visuals and text to summarize findings

Slicers (filters) for year, product category, and territory were added to improve interactivity.

6. Key Insights

- Revenue is heavily driven by a small number of top-performing products
- Product categories contribute unevenly, with dominant categories generating most revenue
- The United States and a few key regions contribute the largest share of sales
- Customer purchasing patterns indicate strong potential for repeat business
- Sales trends show fluctuations over time, suggesting seasonal demand patterns

7. Assumptions

- TotalDue represents total revenue per order
- LineTotal represents product-level revenue
- OrderDate is accurate for time analysis
- CustomerID uniquely identifies each customer
- Missing and duplicate data were removed during cleaning

8. Recommendations and Conclusion

The company should focus on expanding high-performing products and regions while improving underperforming territories. Customer retention strategies such as loyalty programs should be implemented to increase repeat purchases. Additionally, sales trends should be used for better inventory and operational planning.

In conclusion, the analysis provides clear insights into sales performance, customer behavior, and regional trends. Power BI enabled effective visualization and supports informed business decisions.